Theory Concepts & Definitions

PR. 1 — Data Profiler: Customer Purchase Behavior & Churn

Data Preprocessing & Feature Engineering Practical Exam

Part A — Fundamentals

1. What is Data Analysis?

Data Analysis is the process of inspecting, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making.

It typically involves collecting raw data, cleaning it to remove errors and inconsistencies, exploring it to find patterns and relationships, and interpreting the results to answer a business question — in this project, "which customers are likely to churn, and why?"

2. How to Plan a Data Science Project

A structured Data Science project generally follows these steps:

1. Problem Definition — understand the business goal (e.g., predict customer churn).
2. Data Collection — gather data from all relevant sources (CSV, JSON, databases, APIs).
3. Data Understanding — explore structure, types, and quality of the data.
4. Data Cleaning & Preprocessing — handle missing values, duplicates, inconsistent types, irrelevant columns.
5. Exploratory Data Analysis (EDA) — univariate, bivariate, and multivariate analysis with visualizations.
6. Feature Engineering — create or transform features that better represent underlying patterns.
7. Model Building — select and train a suitable ML algorithm on the processed data.
8. Evaluation — assess model performance with appropriate metrics.
9. Deployment & Monitoring — put the model into production and track performance over time.
10. Communication — report insights and results to stakeholders.

3. Machine Learning Problem Statement

Predict whether a customer will churn (Churn: Yes/No), based on their purchase behavior — including total purchases, average order value, total spend, income, support interactions, satisfaction score, and membership details.

This is a binary classification problem. The target variable is Churn, and the goal is to build a model that flags high-risk customers before they leave, so the business can intervene with offers, outreach, or loyalty perks.

4. Tensors

A Tensor is a generalized, multi-dimensional array of numbers — the core data structure used in machine learning and deep learning frameworks such as NumPy, TensorFlow, and PyTorch.

Rank	Name	Example
0-D	Scalar	A single number, e.g. 5

1-D	Vector	A list of numbers, e.g. one customer's [Age, Income, Purchases]
2-D	Matrix	A table of numbers, e.g. multiple customers x multiple features
3-D+	N-D Tensor	A stack of matrices, e.g. batches of customer feature-matrices

In-depth Explanation with NumPy Examples

- A scalar (0-D tensor) is a single value with no axes — e.g., one customer's purchase count.
- A vector (1-D tensor) is an ordered list of numbers along a single axis — e.g., one customer's [Age, Income, Total_Purchases].
- A matrix (2-D tensor) has rows and columns — e.g., 5 customers x 4 numeric features. This is the shape most tabular ML data takes.
- A 3-D tensor stacks multiple matrices — e.g., two batches of 5 customers x 2 features, useful when data has a natural grouping (time steps, batches, channels).

Tensors matter because every ML/DL model operation — dot products, matrix multiplication, gradient updates — is fundamentally a tensor operation. Understanding tensor shape and rank is essential before feeding tabular or image data into a model.

Supporting Concepts Used in the Practical Work

Data Acquisition from Multiple Sources

Real-world datasets rarely live in a single file. This project combines four source formats:

Source	Format	Typical Access Method
Transaction records	CSV	pandas.read_csv()
Customer profile	JSON	json.load() + pd.DataFrame()
Support & satisfaction	SQL (SQLite)	sqlite3 connection + pd.read_sql()
Region enrichment	REST API	requests.get() (with a cached fallback used here, since this exam environment has no outbound network access)

Data Cleaning

- Handling missing data: numeric columns are commonly imputed with the median (robust to outliers); categorical columns with the mode.
- Removing duplicates: exact duplicate rows add no new information and can bias model training.
- Correcting inconsistent types: e.g., converting date strings to proper datetime objects, standardizing inconsistent category labels.
- Dropping irrelevant columns: constant or non-informative columns add noise without predictive value.

Exploratory Data Analysis (EDA)

Analysis Type	Definition	Example in this Project
Univariate	Examines a single variable's distribution	Distribution plots of Age, Income, Total Purchases
Bivariate	Examines the relationship between two variables	Gender vs Purchases, Income vs Churn
Multivariate	Examines relationships among three or more variables	Correlation heatmap, pair plots colored by Churn

Data Profiling

A Data Profiling report is an automated summary of a dataset's structure and quality, typically covering: row/column counts, missing values, descriptive statistics, correlations between features, and data-quality warnings (e.g., high skew, high missingness, near-duplicate/high-cardinality columns). Tools like ydata-profiling automate this; this project implements an equivalent custom function so the report can run in an offline, package-restricted environment.

Summary

Together, these concepts — data analysis, project planning, ML problem framing, tensors, multi-source acquisition, cleaning, EDA, and profiling — form the foundation of turning raw, messy, multi-format data into a clean, well-understood dataset that is genuinely ready for machine learning.